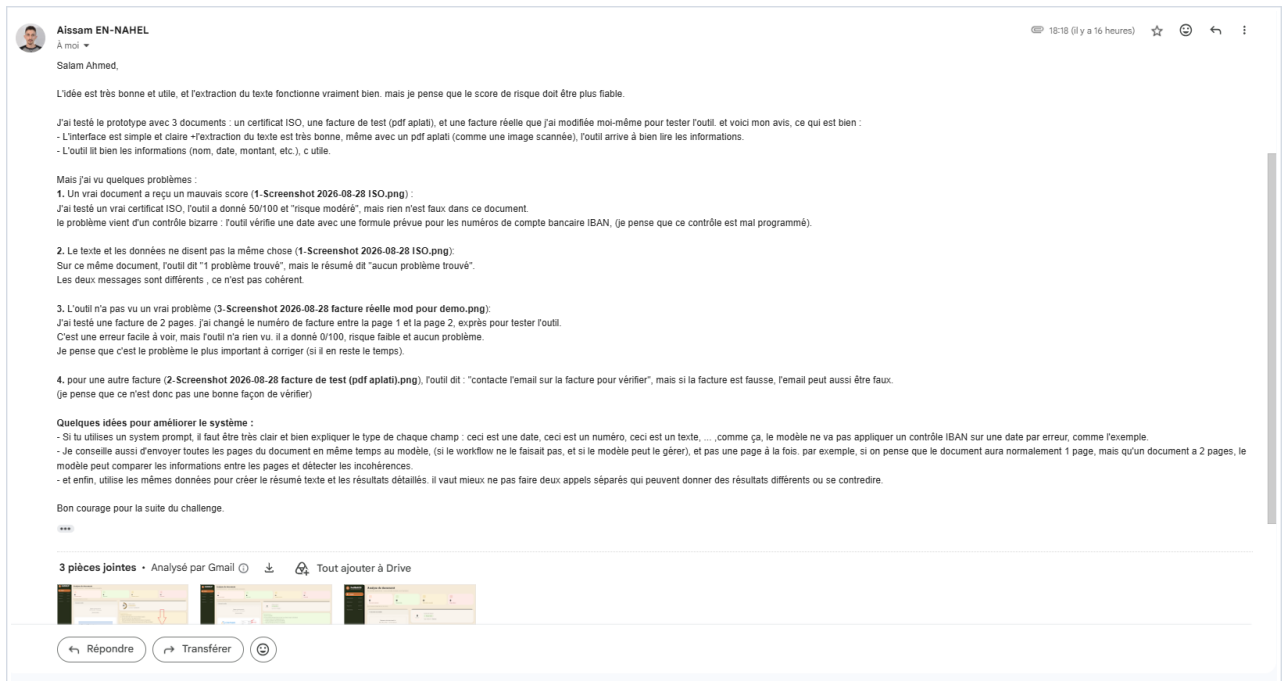


1. First external validation — TecForge

Company	TecForge
Tester	Aissam EN-NAHEL, Human Resources — written feedback, 28 August 2026
Material	Three documents: a genuine ISO certificate, a flattened test invoice, and a real two-page invoice the tester altered himself to see whether the tool would notice.
Method	Unassisted use of the deployed prototype, followed by written feedback. No script, no guidance.

Verdict in the tester's own words

The idea was judged useful and the text extraction very good — including on a flattened PDF read as a scanned image. *“Je pense que le score de risque doit être plus fiable”*: the concern was the reliability of the risk score, not the reading of the document.



Feedback received from Aissam EN-NAHEL (TecForge), reproduced verbatim.

Held up under test

- ✓ Interface judged simple and clear.
- ✓ Extraction accurate even on a flattened PDF.
- ✓ Fields read correctly: name, date, amount.

Did not hold up

- ✗ A genuine document scored as suspicious.
- ✗ Narrative contradicted the anomaly count.
- ✗ A deliberate forgery went unnoticed.

2. The four defects, and their cause in the code

Each report was traced to a specific line before anything was changed. None was a tuning problem; all four were logic errors.

- 1. A genuine ISO certificate scored 50/100.** The IBAN checksum fired on the shape of a value while ignoring its label. “Two letters, two digits, then 11 to 30 characters” describes a certificate number as well as a bank account, so the mod-97 key failed on an authentic document — and a single failed check imposed a score floor of 50, turning a false positive into “moderate risk”.
- 2. The count said one problem, the summary said none.** The model writes its explanation before the deterministic checks run. It could therefore report “nothing found” while those checks were adding an anomaly to the list. Two true statements about two different stages, printed as one contradiction.
- 3. An invoice number changed between page 1 and page 2 went unseen.** The PDF renderer read `getPage(1)` only, while reporting the real page count. The altered page never reached the model. The tester ranked this the most important defect; the diagnosis confirms it, since no amount of prompt tuning could have recovered a page that was never sent.

4. **“Contact the email on the invoice to verify.”** A forged document carries forged contact details. The prompt asked for “concrete verification steps” without ever forbidding the document from vouching for itself.

3. From feedback to shipped change

Every point raised was corrected and deployed within the challenge window, not deferred to a backlog. The changes are public in the project repository: github.com/Hajji-ahmed/TrustDoc-AI.

FIXED means the correction is written, tested by assertion and deployed. **PROVEN** means it was afterwards observed working in a run produced by someone else. **RETEST PENDING** means the fix is live but the case that exposed the defect has not been run again. The three are never merged.

Reported	Change made	Status
Wrong rule applied to a date	The IBAN check now requires a label that names a bank account. An IBAN filed under a label that does not say so escapes the check — accusing a genuine document costs more than missing one verification.	FIXED
Summary contradicts detail	Each statement is attributed to its stage: visual analysis first, automatic checks second. The model’s prose is kept, not rewritten. Later observed in the second reviewer’s own output.	PROVEN
Two pages never compared	All pages are rendered and sent to the model in one request, each announced by its number. The prompt requires values repeated across pages to be compared. Pipeline verified; the model’s ability to spot the difference is not.	RETEST PENDING
Email alone is weak evidence	The prompt forbids any verification routed through the document’s own contact details, and requires an independent source instead. Later observed in the second reviewer’s own output.	PROVEN
Field types left implicit (tester’s suggestion)	The prompt now states which labels may be used for which value types, so the wrong deterministic check cannot be triggered downstream.	FIXED
Cross-page detection on the tester’s own file	Route contract verified on six cases; the two-page invoice has not been run again through the model.	RETEST PENDING

What this round actually taught

Extraction quality was never the weak point — the tester confirmed it works, including on flattened scans. Every defect he found sat in the layer that decides *what the reading means*: which rule applies to which field, whether pages are compared, whether the narrative matches the findings. A screening tool is not limited by what it can read, but by the discipline of what it concludes.

The correction that has not yet been proven

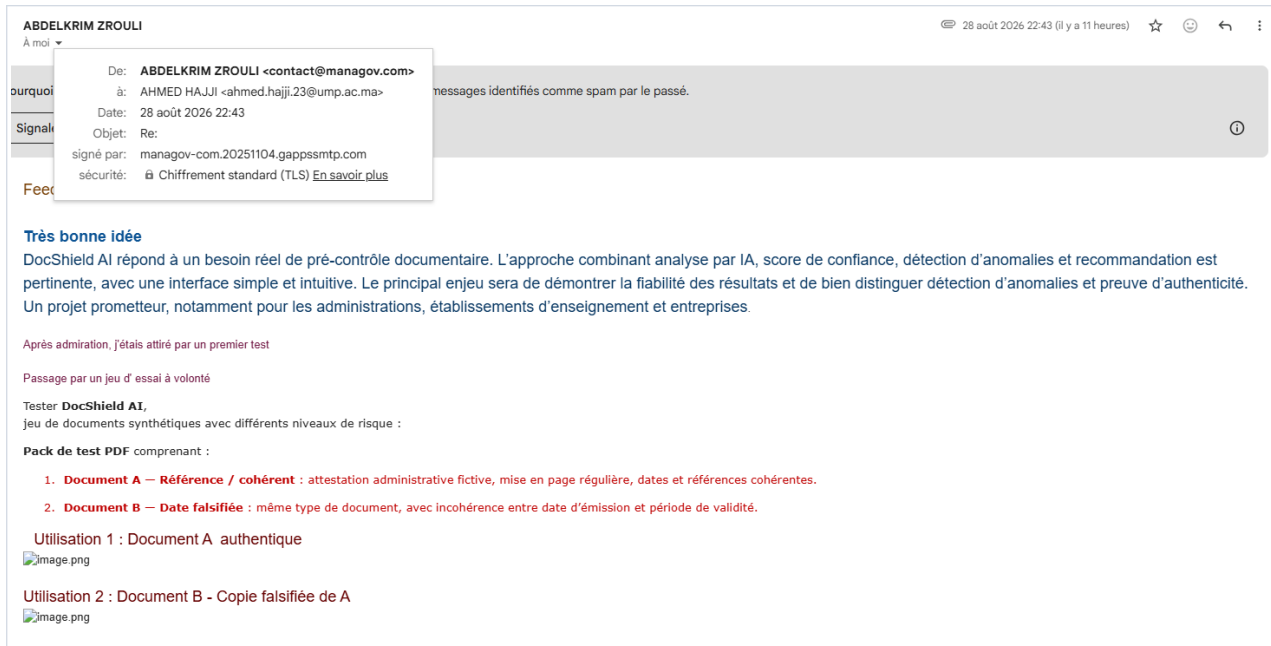
Points 1, 2 and 4 are verified by automated assertions. Point 3 — the one the tester ranked highest — is verified only up to the request contract: all pages now reach the model. Whether the model *notices* the altered invoice number requires re-running his own two-page file. Until that is done, this defect is fixed in the pipeline and unproven in the result.

4. Second external validation — Managov

Company	Managov
Reviewer	Abdelkrim Zrouli, Director — written feedback, 28 August 2026
Material	A synthetic test pack he built himself: Document A, a coherent administrative attestation, and Document B, the same document with its issue date made inconsistent with its validity period.
Method	Unprompted assessment of the concept, followed by a controlled A/B test on a matched pair of documents.

Assessment

“DocShield AI répond à un besoin réel de pré-contrôle documentaire.” The combination of AI analysis, confidence score, anomaly detection and recommendation was judged relevant, the interface simple and intuitive, and the project promising for public administrations, educational institutions and companies.



Feedback received from Abdelkrim Zrouli (Managov), reproduced verbatim.

The reservation, and why it matters more than the praise

“Le principal enjeu sera de démontrer la fiabilité des résultats et de bien distinguer détection d’anomalies et preuve d’authenticité.”

This is the product’s own founding constraint, stated back by someone who had not been told it. DocShield never claims a document is authentic; it reports what looks irregular and hands the decision to a human. Hearing that boundary named as *the* central challenge, by a director evaluating the tool for his own organisation, confirms that the distinction is not academic — it is what buyers will judge the product on.

4.1 The test design — one controlled variable

Both documents are byte-for-byte identical except for the validity window. That is a cleaner experiment than anything run in-house: it isolates a single failure mode and ships with its own control.

Field	Document A — control	Document B — altered
Issue date	28 August 2026	28 August 2026
Period covered	1 Jan 2026 to 31 Dec 2026	1 Jan 2026 to 31 Dec 2025
Validity	until 31 Dec 2026	until 31 Dec 2025
Everything else	Identical — reference, identifier, issuer, signature, verification reference	

The planted defect is a period that ends before it begins, on a document issued after both dates. It is obvious to a human and invisible to any check that reads fields one at a time.

4.2 The result — the planted defect was caught

Document B, scored by the deployed prototype

Score 34/100 — moderate risk. **Type** Attestation administrative, correctly identified.

Anomaly One, and only one: “*La période concernée commence en 2026 mais termine en 2025, ce qui est chronologiquement impossible.*” Classified under data coherence, moderate severity, and drawn as a numbered region on the document itself.

Recommendation Manual verification required — confirm the details with the issuer *through alternative official channels*, and compare against other documents.

What this result confirms beyond the detection itself

Two of the corrections made after the first validation are visible in this output, independently produced by a second reviewer who knew nothing of them.

The narrative no longer contradicts the findings. The explanation reads “*Analyse visuelle: . . .*” then “*Contrôles automatiques: aucun écart relevé. . .*”. Each statement is attributed to its stage, and both are true at once: the model saw the date problem, the deterministic checks correctly found nothing in their own scope.

The verification step no longer routes through the document. The recommendation names *alternative official channels* rather than the contact details printed on the page — the precise flaw reported by the first reviewer.

The control was not scored

Document A exists to answer the question the reviewer wrote at the top of both files: “*Vérifier que DocShield AI ne génère pas de faux positifs.*” It was uploaded, but the captured screen shows it before the analysis was run, so **no result is recorded for the control**.

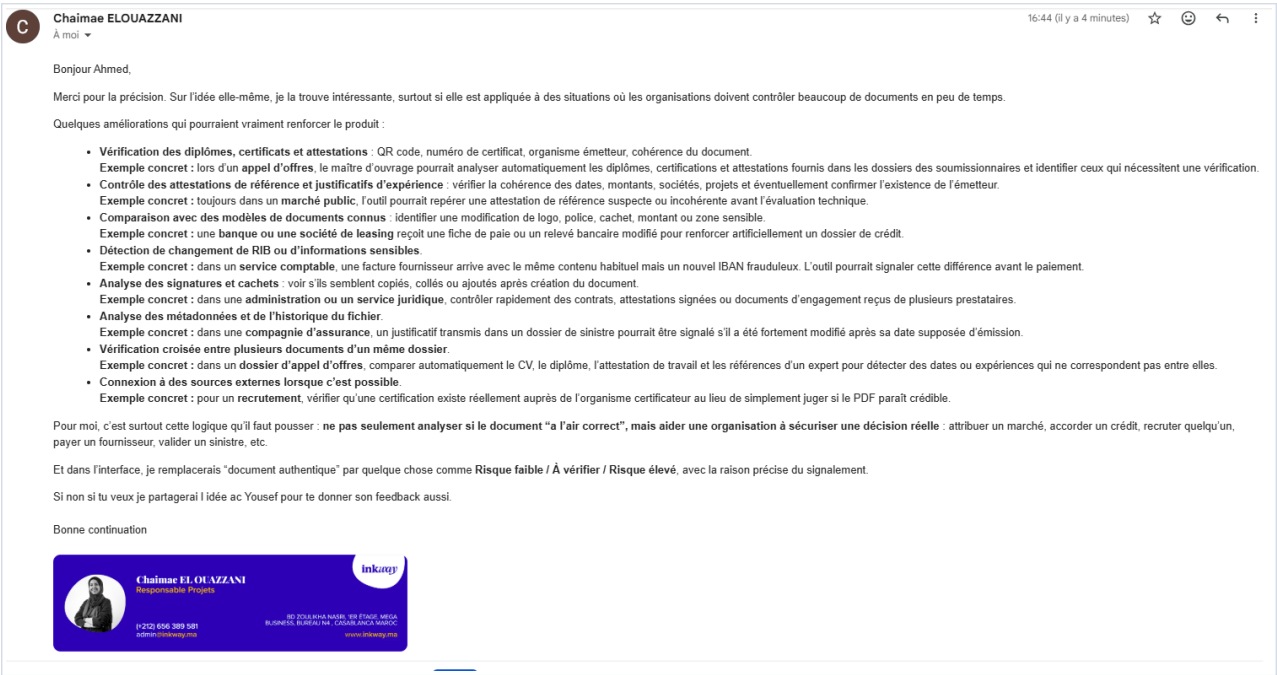
Detection on B is therefore established; the absence of a false positive on A is not. A test that only ever fires proves nothing on its own — the control is what separates a detector from an alarm that is always on. Running Document A is the single most useful next step in this dossier.

5. Third external validation — INKWAY

Company	INKWAY
Reviewer	Chaimae EL OUAZZANI, Project Manager — written feedback, 29 August 2026
Material	No document test. An assessment of the product direction, given once the current scope had been explained.
Method	Unprompted review, returned as eight numbered proposals, each with the operational situation it would serve.

Where the idea earns its place

The concept was judged useful “surtout si elle est appliquée à des situations où les organisations doivent contrôler beaucoup de documents en peu de temps”. Volume under time pressure is the qualifying condition — the second reviewer to reach that criterion independently, and the same reasoning that put lending and leasing at the head of deliverable 04.



Feedback received from Chaimae EL OUAZZANI (INKWAY), reproduced verbatim.

5.1 The eight proposals, against what exists

Recorded as a roadmap rather than a wish list, each measured against the prototype as it stands today.

Proposed	Against the prototype	Status
Diplomas and certificates: QR code, certificate number, issuing body	Internal coherence is checked. Reading a QR code and looking up an issuer are not.	NOT BUILT
Reference attestations: dates, amounts, companies, existence of the issuer	Dates and arithmetic are checked. Confirming that the issuer exists requires an external source.	PARTIAL
Comparison against known document templates	The model judges plausibility, never conformity to a reference model. It has no template to compare with.	NOT BUILT
Change of bank details or other sensitive fields	The IBAN check key is verified. Detecting a <i>change</i> requires knowing the supplier’s usual details — a history the tool does not keep.	PARTIAL
Signatures and stamps: copied, pasted, added after the fact	The model is asked to inspect seals and signatures visually. No forensic analysis.	PARTIAL
Metadata and file history	Excluded by construction: the document is rasterised before analysis, so metadata never reaches the model.	NOT BUILT
Cross-checking several documents of one file	Cross-page comparison within a single document ships today. Cross-document does not.	PARTIAL
Connection to external sources	Not built. This is the boundary the second reviewer named, reached here from a different direction.	NOT BUILT

The reframing, which matters more than any single item

“Ne pas seulement analyser si le document a l’air correct, mais aider une organisation à sécuriser une décision réelle” — awarding a contract, granting credit, hiring, paying a supplier, settling a claim.

That moves the unit of value from the document to the decision it supports. Six of the eight proposals follow from it: they are not detection features, they are what a reviewer needs before signing. It also explains why external sources recur — a decision cannot be secured by a document vouching for itself.

On the interface The reviewer would replace any wording suggesting *authentic* with *low risk / to be checked / high risk*, plus the precise reason for the flag. The risk badge already reads that way; the recommendation label “*Document acceptable*” does not, and still implies a verdict on authenticity. Small change, and directly on the boundary all three reviewers have now named.

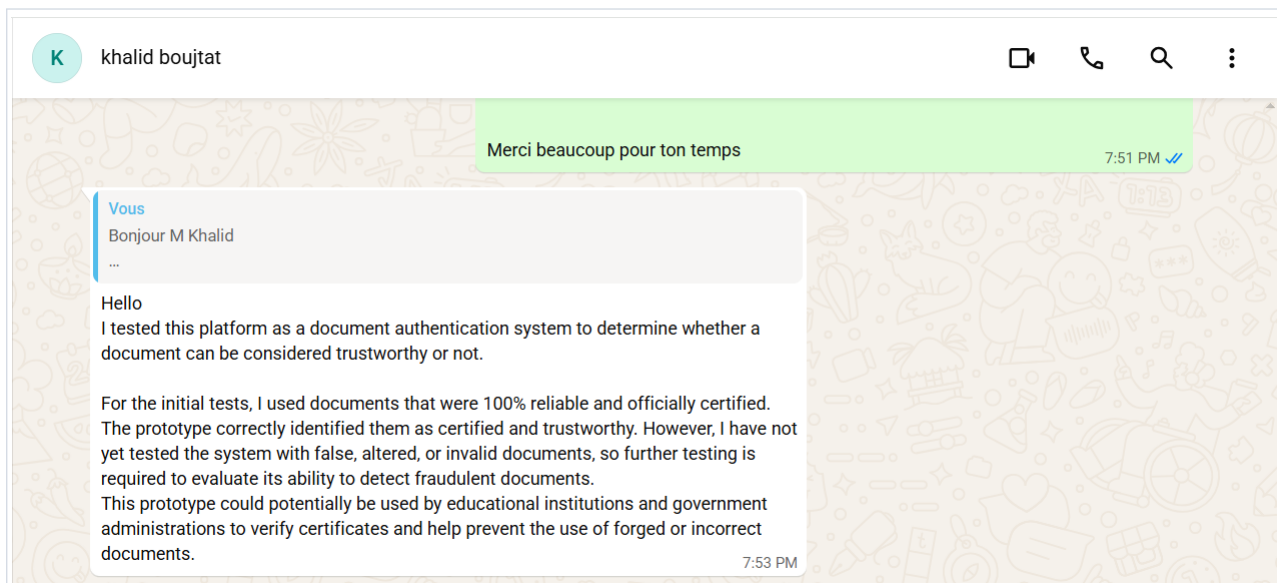
What three independent reviews converge on

An HR practitioner, a company director and a project manager, none aware of the others’ feedback, arrived at the same boundary from three directions: the score must be trustworthy, detection must stay distinct from proof, and the interface must never say *authentic*.

No reviewer asked for more detection categories. All three asked for the product to be honest about what it does not establish. That is the clearest product signal the challenge produced.

6. Fourth external validation — Actemium

Company	Actemium
Reviewer	Khalid Boujtat, Electrical Engineer — written feedback, 29 August 2026
Material	Officially certified documents, described by the reviewer as fully reliable. No altered or invalid document was submitted.
Method	Unassisted use of the deployed prototype, with the stated role of the tool as the only briefing.



Feedback received from Khalid Boujtat (Actemium), reproduced verbatim.

The evidence the dossier was missing

Genuine, officially certified documents went through the prototype and came back unflagged. This is the first independent confirmation that the tool does not raise a false positive on real documents — the exact gap left open by the second validation, whose control document was never scored.

The reviewer states his own limit without being asked: no false, altered or invalid document was tested, so his run says nothing about detection. Detection and absence of false positives are now each established by a different reviewer, and neither by the same one.

The wording problem, no longer an opinion

The reviewer describes the tool as a “*document authentication system*” and reports that it “*correctly identified them as certified and trustworthy*”.

The prototype certifies nothing. It reported an absence of detected anomalies, and a qualified engineer read that as certification. The third reviewer had recommended dropping any wording that suggests authenticity; this run shows the misreading actually happening, on someone competent, with no ambiguity of language. That turns an interface recommendation into a demonstrated defect — and it is the one finding in this dossier that a reviewer produced without intending to.

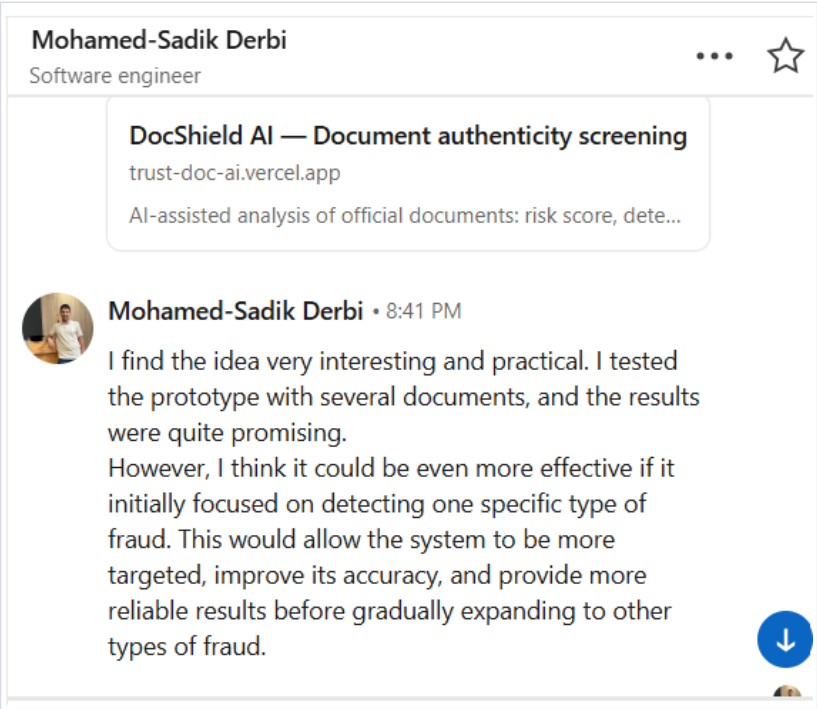
Second independent convergence on the same segments

Asked nothing about markets, the reviewer proposes **educational institutions and government administrations** for certificate verification — the same two segments the second reviewer named on his own initiative, from a different company and a different role.

Two unprompted mentions do not make a market study, but they are two more than the beachhead in deliverable 04 has received.

7. Fifth external validation — Verve Technologie

Company	Verve Technologie
Reviewer	Mohamed-Sadik Derbi, Software Engineer — written feedback, 29 August 2026
Material	Several documents, run against the deployed prototype.
Independence	Partial, and stated here rather than left to be discovered: this reviewer also appears in deliverable 05 as a proposed team member. His opinion is informed, not unconnected.



Feedback received from Mohamed-Sadik Derbi (Verve Technologie), reproduced verbatim.

The recommendation: go deep before going wide

The idea was judged interesting and practical, the results “quite promising”. The reservation is one of scope: the prototype would be more effective “if it initially focused on detecting one specific type of fraud”, gaining accuracy and reliability before expanding to others.

This contradicts the third review — and the contradiction is useful

The third reviewer returned eight capabilities to add. The fifth says add none yet, and narrow instead. Both are experienced; both cannot be followed at once. They disagree on the route, not the diagnosis. Each is reacting to the same weakness — the tool is general and therefore shallow. One reads that as missing capabilities, the other as insufficient depth. The eight proposals describe what a decision eventually needs; the fifth review describes the order in which to get there.

Where the dossier says to start, if one type is chosen

The evidence already points to one. The only falsification the prototype has been shown to catch, on a document designed by someone else, was an internal date inconsistency on an administrative attestation. Cross-page consistency ships today. And two reviewers, unprompted and from different companies, named education and public administration — both certificate-heavy, both document families where dates, periods and reference numbers carry the meaning. Narrowing to **certificates and administrative attestations**, judged on internal data consistency, would follow the fifth reviewer’s advice without discarding anything already built or proven.

8. Validation register

One line per company contacted during the challenge. Only verified conversations and real feedback are recorded here; the blank rows are reserved for the validations still in progress.

Company	Person / Role	Feedback / Learning	Change / Next action
TecForge	Aissam EN-NAHEL <i>Human Resources</i>	Document reading judged reliable, including on flattened scans; the risk assessment was not. Four defects reported: a validation rule applied to the wrong type of field, a summary inconsistent with the detailed findings, no comparison between the pages of a single document, and a verification step relying on the document's own contact details.	All four defects corrected and deployed within the challenge window. Next: retest the altered two-page invoice, then extend testing to other document families.
Managov	Abdelkrim Zrouli <i>Director</i>	Concept and interface judged relevant for administrations, schools and companies. Central challenge named: demonstrating the reliability of results, and keeping anomaly detection clearly distinct from proof of authenticity. Ran a matched pair of synthetic documents differing by one falsified validity window; the altered document was correctly flagged.	Positioning confirmed and kept explicit in the interface. Detection validated on the altered document. Next: run the control document to confirm no false positive is raised.
INKWAY	Chaimae EL OUAZZANI <i>Project Manager</i>	Concept judged useful where organisations must check many documents under time pressure. Eight proposals returned, each tied to an operational situation, all pointing the same way: secure a decision, not score a document. Would drop any wording suggesting authenticity in favour of a risk level with its reason.	Proposals recorded against the current state as a roadmap. Interface wording to revise. Volume under time pressure confirmed as the qualifying condition.
Actemium	Khalid Boujtat <i>Electrical Engineer</i>	Officially certified documents went through unflagged — first confirmation of no false positive on genuine documents. Detection untested by him, and he says so. Read the result as the document being “certified and trustworthy”, which the tool never claims. Proposes education and public administration.	Absence of false positives now evidenced. Interface wording confirmed as a real defect, not a preference. Next: have him run an altered document.
Verve Technologie	Mohamed-Sadik Derbi <i>Software Engineer</i>	Idea judged practical, results promising on several documents. Would narrow the prototype to one type of fraud first, for accuracy and reliability, before widening. Also listed in deliverable 05 as a proposed team member, so informed rather than unconnected.	Sequencing question opened against the third reviewer's eight proposals. Candidate first vertical identified from existing evidence: certificates and administrative attestations.

9. Every test, and exactly what it established

Five reviewers did five different things. Presenting them as five equivalent technical validations would overstate the dossier, so the nature of each is named.

Test	Result	Status
Altered administrative attestation, designed by the reviewer Managov — controlled test	The planted date inconsistency was detected, named and localised.	PROVEN
Control document of that same pair Managov — controlled test	Uploaded, but the captured screen predates the analysis. No result recorded.	RETEST PENDING
Officially certified documents Actemium — independent technical test	Passed unflagged. Evidence of no false positive on the samples tested.	PROVEN
Wrong validation rule on a date field TecForge — independent technical test	False positive on a genuine ISO certificate. Rule now requires a matching label; covered by assertions.	FIXED
Narrative contradicting the findings TecForge — independent technical test	Fixed, and afterwards observed corrected in another reviewer's output.	PROVEN
Two-page invoice with an altered number TecForge — independent technical test	Root cause fixed and pipeline verified. The file itself has not been re-run through the model.	RETEST PENDING
Verification routed through the document TecForge — independent technical test	Prompt now requires an independent source; observed working in another reviewer's output.	PROVEN
Well-made or AI-generated document Reported by a reviewer	Passes every check. A coherent fabrication is invisible to screening — a limitation, stated rather than hidden.	NOT BUILT
Product direction and scope INKWAY — product feedback, no document test	Eight capability proposals, each tied to an operational situation. No detection claim follows from it, and none is made.	PARTIAL
Several documents, then a scope objection Verve — test plus product judgement	Results judged promising; no specific defect reported. The contribution is the sequencing argument, not a measured result.	PARTIAL

10. Summary

Completed Five external companies, five reviewers — an HR practitioner, a company director, a project manager, an electrical engineer and a software engineer — four defects found and four corrections shipped within the challenge window. Four of the five had no prior connection to the project.

Confirmed from outside The boundary the product was built on — report anomalies, never certify authenticity — was reached independently by every reviewer; none of them told it and none aware of the others. Two of the corrections made after the first validation are visible in the second reviewer's test output. No reviewer asked for more detection categories.

Detection demonstrated On a falsification designed by a third party, the prototype isolated the planted defect, named it correctly and localised it on the document.

Absence of false positives demonstrated Officially certified documents, submitted by a fourth reviewer, went through unflagged. The two properties are evidenced by two different people, which is stronger than one reviewer reporting both.

Confirmed as a defect The interface lets its output be read as certification. The third reviewer predicted it; the fourth did it.

Open question Scope. One reviewer would add eight capabilities, another would narrow to a single type of fraud first. The disagreement is about order, not direction, and the evidence points to certificates and administrative attestations as the first vertical.

Next steps Have the fourth reviewer run an altered document, so detection and its absence come from the same hand. Re-run the two-page invoice from the first validation. Revise the wording that suggests authenticity.